CHANDU PRASAD BHAIRAPU



GenAI Developer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in generative AI solutions and automation for business processes, from LLMs, RAG and AI agents to API integration and the step from proof of concept to deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by AI consulting work with business units at Covestro and a strong foundation in Python and SQL.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end, from proof of concept to a handed-over system: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized AI agents for document understanding (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph through an API for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, letting domain experts validate and approve knowledge-graph changes.
- Applied Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, Neo4j, MongoDB, Postgres, SQL, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business units to understand their decision-making workflows and advised where agentic AI could support them; designed and deployed an agentic LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a Python backend with FastAPI APIs and a React frontend, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React, Postman**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data to choose the retrieval setup.

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Azure AI Search, Vector Databases, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Order Inbox Agent | Microsoft Graph | SharePoint | LiteLLM | Ollama | pgvector

- Order-intake automation: purchase orders from a Microsoft 365 mailbox (Graph API) are filed in a SharePoint DMS, extracted by an LLM, matched to the product catalog with pgvector and written to a mock ERP via REST API; a labelled order set compares a cloud LLM with an open-source model (Ollama) on accuracy, cost and latency.

TECHNICAL SKILLSET

- **Generative AI:** Large Language Models, Retrieval-Augmented Generation (RAG), GraphRAG, AI agents (LangGraph, LangChain), LiteLLM, prompt engineering, MCP, open-source LLM fine-tuning (QLoRA), RAGAS
- **Software & Integration:** Python, SQL, FastAPI, REST APIs, system integration via APIs, Python microservices, React (beginner), Postman, Git, GitLab CI/CD, Agile
- **Data & Cloud:** Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Azure, Docker, Kubernetes
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India